

WHO KNOWS... THE GAME IS THE GAME

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Abstract

Declaration

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List of Abbreviations

AC Conditional Addition. 1, 43, 49

AMP Automatic Mixed Precision. 1, 40

ANN Artificial Neural Network. 1, 25

ATan ArcTangent. 1, 26

BAF Background Activity Filtering. 1, 18

CNN Convolutional Neural Network. 1, 31–34, 40, 42–49

CPU Central Processing Unit. 1, 47

ETraM Event-based Traffic Monitoring Dataset. 1, 16, 18, 21, 25, 35–37, 49

FLOP Floating-Point Operation. 1, 42, 48

FPS Frames Per Second. 1

FRR Firing Rate Regularization. 1, 17, 40

GPU Graphics Processing Unit. 1, 47–49

HD High Definition. 1, 18

HPC High Performance Computing. 1, 17, 20, 22

IoU Intersection over Union. 1, 12, 21, 34, 37–39, 41, 44

LIF Leaky Integrate-and-Fire. 1, 2, 17, 20, 25–27, 40, 43, 46, 47

LRU Least Recently Used. 1, 23

MAC Multiply-Accumulate. 1, 42, 43, 49

mIoU Mean Intersection Over Union. 1, 44, 45, 49

ReLU Rectified Linear Unit. 1, 25

ResNet Residual Neural Network. 1, 27

SE Squeeze-and-Excitation. 1, 29, 34

SEW Spike-Element-Wise. 1, 5, 28

SNN Spiking Neural Network. 1, 17–19, 25–27, 29–35, 37, 40, 42–49

SNR Signal-to-Noise Ratio. 1, 18, 35

SOP Synaptic Operation. 1, 18, 41, 43, 48, 49

Chapter 1

Technical Quality, Methodology & Evaluation

1.1 Requirements Analysis

The primary objective of this project was the development and rigorous evaluation of a high-performance Spiking Neural Network (SNN) for multi-object detection using high-definition event-camera data (the ETraM dataset). To effectively quantify the accuracy versus energy efficiency tradeoff between bio-plausible spiking architectures and traditional Convolutional Neural Networks (CNNs), the system had to be designed against a stringent set of functional and non-functional requirements. These requirements guided the iterative evolution of the neural architectures and the underlying computational pipeline.

1.1.1 Functional Requirements

To ensure a fair empirical comparison and to solve the complexities of multi-object detection on neuromorphic data, the following functional criteria were established:

- **Multi-Object Regression Capabilities:** Unlike simplified classification tasks, the system was required to perform multi-object detection capable of identifying multiple vehicles simultaneously within a single scene. This necessitated a regression head capable of outputting multiple bounding boxes per frame (defined by spatial coordinates and confidence scores) while actively suppressing “ghost detections” and resolving spatial ambiguities.

- **Architectural Parity (The “Gold Standard” Baseline):** To strictly isolate the impact of the computational paradigm (event-driven spiking versus continuous-valued activations), the baseline CNN was required to be a 1:1 non-spiking counterpart to the SNN. Both models were constrained to equivalent structures, depths and parameter counts ($\sim 11.7\text{M}$ parameters). Crucially, to eliminate data variance as a confounding factor, both the SNN and the CNN were trained on the exact same pre-processed neuromorphic dataset (the REPLICA pipeline output). This ensured that any observed discrepancies in performance or power consumption were solely attributable to the underlying network architectures, rather than differing input modalities or varied model capacity.
- **High-Definition Spatiotemporal Ingestion:** The system was required to process raw, high-definition (1280×720) event streams. Furthermore, the pipeline had to preserve the temporal dynamics of the data by segregating it into discrete time bins (e.g., $T = 10$) to allow the SNN to demonstrate temporal evidence accumulation.
- **Comprehensive Evaluation Metrics:** The evaluation suite was required to output granular metrics beyond top-line Mean Intersection over Union (IoU). This included real-time dynamic power consumption mapping, confidence calibration analysis (reliability diagrams), and quantization error tracking across varying object sizes.

1.1.2 Non-Functional Requirements & Constraints

The processing of high-definition event data at scales of up to 150 training epochs imposed severe computational bottlenecks. Consequently, the system had to satisfy critical performance and stability constraints:

- **Computational Scalability & Speed:** The SNN simulation had to be optimized for execution on local accelerator hardware (GPUs). This required the integration of Automatic Mixed Precision (AMP) to utilize Tensor Cores, memory-efficient vectorized loss functions to eliminate sequential batch loops, and the utilization of optimized C++/CUDA kernels (e.g., SpikingJelly’s multi-step modes) over native Python loops.
- **Numerical Stability:** Given the reliance on 16-bit precision through AMP and the deep accumulation of membrane potentials in the SNN, the architecture required

specific safeguards against gradient explosion. This constrained the design of the detection head, mandating the use of raw logits and numerically stable loss functions (e.g., `BCEWithLogitsLoss`).

- **Hardware Extrapolation:** While constrained to local GPU execution, the benchmark outputs (e.g., dynamic power tracking) were required to be translatable into theoretical energy consumption metrics (in mJ) based on established hardware constants, facilitating projections for specialized neuromorphic chips like Intel Loihi or IBM TrueNorth.

These requirements formed the foundation for the three-generation evolution of the `SNN_MAX_REPLICA_SE` architecture detailed in Section 1.4, and drove the rigorous benchmarking suite evaluated in Section ??

1.2 System Architecture Overview

To satisfy the stringent requirements established in Section 1.1, the overall system was designed as an end-to-end, highly modular pipeline. The modularity of this pipeline was a deliberate methodological choice to facilitate systematic design exploration. By decoupling the preprocessing, neural backbone, and regression stages, we enabled the isolation and empirical study of individual architectural modifications, ensuring that comparative findings between the spiking and continuous paradigms remained robust and controlled. The high-level data flow and structural components of the system are formalized in the block diagram presented in Figure 1.1.

As illustrated, the architecture is divided into four primary sequential stages, where the central neural backbone acts as an interchangeable module:

1. **Data Ingestion & The REPLICA Pipeline (Fixed):** Raw, asynchronous event data at a high spatial resolution of 1280×720 is ingested from the ETraM dataset. Due to the conventionally sparse and noisy nature of neuromorphic hardware outputs, the data enters the *REPLICA* preprocessing pipeline, which is heavily adapted from the methodology proposed in the original ETraM study Verma et al. 2024. While the core spatiotemporal noise filtering and bilinear temporal interpolation (to $T = 10$ bins) remain faithful to the original study, this project included an additional spatial downsampling stage ($ds = 4 : 1280 \times 720 \rightarrow 320 \times 180$). This modification was strategically implemented to ensure

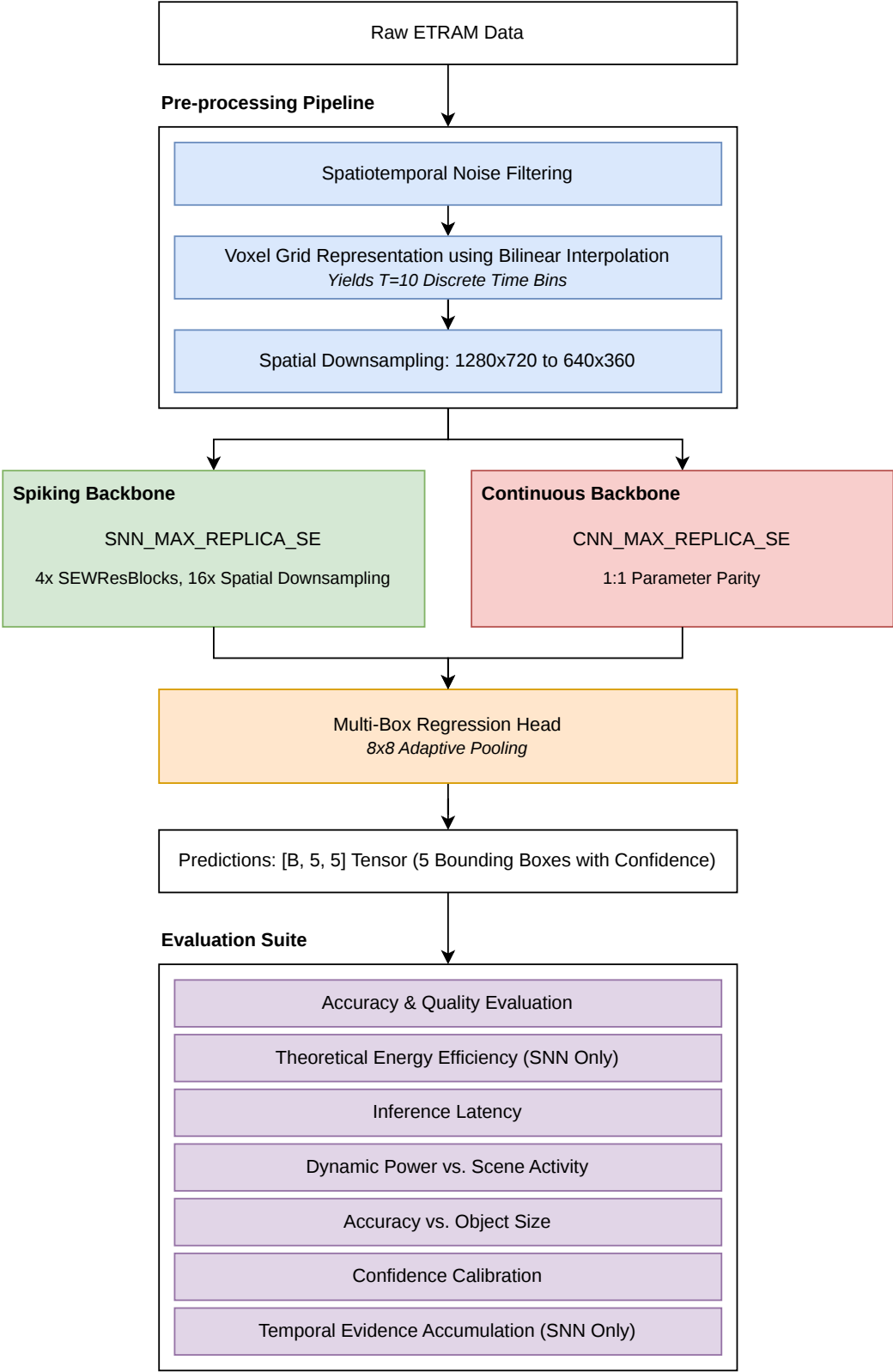


Figure 1.1: The high-level system architecture pipeline, illustrating the flow from raw ETRaM high-definition event data, through the fixed REPLICA preprocessing stages, into the interchangeable neural backbone (SNN or CNN), and concluding with the multi-box regression head and the comprehensive evaluation suite.

training feasibility on local accelerator hardware and reduce input sparsity, thereby improving the stability of neuronal feature extraction. This stage is detailed thoroughly in Section 1.3.

2. **The Neural Backbone (Interchangeable):** The preprocessed output of the *REPLICA* pipeline serves as the input to the primary feature extraction engine. This stage houses either the spiking architecture `SNN_MAX_REPLICA_SE` or its exact continuous counterpart, `CNN_MAX_REPLICA_SE`. Both backbones utilize four functional blocks incorporating Squeeze-and-Excitation (SE) mechanisms and Residual connections (`SEWResBlock`) to extract deep semantic features while performing an aggressive $16\times$ total spatial downsampling. The evolution and formal design of these backbones are detailed in Section 1.4.
3. **Multi-Box Regression Head (Fixed):** The final abstracted feature maps from either backbone are fed into a unified detection head. To resolve the spatial ambiguities and “ghost detections” observed in earlier architectural iterations, the head utilizes an 8×8 Adaptive Pooling mechanism to aggregate features from 64 spatial receptive blocks. This abstracted information is then mapped to a final output tensor of shape $[B, 5, 5]$, representing the top 5 predicted bounding boxes per image. Each detection is defined by the feature set $\mathcal{F} = \{x, y, w, h, c\}$, where (x, y, w, h) denote the normalized bounding box coordinates and c represents the object confidence logit. The regression output is evaluated via a fully vectorized Multi-Box Loss function during training.
4. **Evaluation & Benchmarking Suite (Fixed):** The predictions and internal neuronal activity from the models are passed into a specialized benchmarking suite. This stage is responsible for the rigorous empirical comparison of the two architectures across five distinct dimensions: real-time dynamic power consumption, accuracy variations relative to object size (addressing the inherent “quantization error” in SNNs Cordone, Miramond and Thierion 2022), model confidence calibration, temporal evidence accumulation, and latency observed on local hardware. These metrics were specifically selected to satisfy the project’s success criteria and provide a comprehensive evaluation of the neuromorphic paradigm. This standardized evaluation framework ensures that the findings are scientifically robust and reproducible. This benchmarking suite is detailed thoroughly in Section 1.7.

Using the modular pipeline guarantees that the SNN and CNN variants are evaluated identically, receiving the exact same preprocessed event volumes and outputting predictions through an identical regression and benchmarking framework.

1.3 Data Preprocessing: The REPLICA Pipeline

Raw neuromorphic event data is fundamentally different from standard frame-based video: instead of capturing absolute intensity at a fixed frame rate, an event camera asynchronously records per-pixel changes in log-intensity, outputting a continuous stream of events in the form $e = (x, y, t, p)$, where (x, y) are the spatial coordinates, t is the microsecond timestep, and $p \in -1, 1$ is the polarity of the event (indicating an increase or decrease in brightness).

While this asynchronous nature offers high temporal resolution and high dynamic range, raw event streams are inherently sparse, noisy and incompatible with conventional deep learning frameworks that expect dense, structured tensors. To bridge this gap, the raw Event-based Traffic Monitoring Dataset (ETraM) data (1280×720 spatial resolution) was passed through a preprocessing pipeline before network ingestion.

1.3.1 Raw Event Ingestion & Windowing

The REPLICA pipeline begins by reading the raw h5 event files (x, y, t, p) and the partnering npy files containing the bounding box annotations from the ETraM dataset. Once both files have been parsed, the first step taken is to apply a $100ms$ sliding window across the entire duration of the event file. “Windowing” is the process of slicing the continuous event stream into chunks that represent a single instance of time for the model, thereby anchoring the asynchronous event stream into discrete, structured training instances.

To maximize training efficiency and ensure a high signal-to-noise ratio in the training distribution, the REPLICA pipeline adopts a Box-Central sampling strategy rather than more conventional uniform temporal partitioning, where training windows were anchored deterministically around the centroids of existing ground truth annotations. This approach ensures that 100% of the training voxels contain informative structural targets, mitigating the prevalence of ‘empty-scene’ noise and providing the SNN with consistent temporal evidence to drive membrane potential accumulation prior to the target timestamp.

In this project, the time window $\Delta T = 100\text{ms}$ was chosen to align with the temporal persistency of the human visual system of roughly 100ms Deodato and Melcher 2024; just as the human eye integrates luminance changes over approximately one frame of perception, the Spiking Neural Network (SNN) also integrates the asynchronous event stream over the same duration before producing a detection output.

The practical consequence of this choice is two-fold. First, a 100ms window is long enough to guarantee a minimum event density sufficient for the LIF neurons in the stem layer to accumulate membrane potential and produce meaningful spike trains across all $T = 10$ temporal bins. Second, specific to the context of vehicle detection, it is short enough that a vehicle travelling at highway speed (e.g., $120 \text{ km/h} \approx 33 \text{ ms}^{-1}$) displaces by only 3.3m within the window, ensuring that the bounding box annotation assigned to the window remains spatially valid and does not introduce label noise through excessive object displacement.

In combination with the Box-Central sampling approach, the pipeline applies a density filter prior to voxelization to enforce a minimum quality threshold in valid samples. For each 100ms window, the total number of raw (x, y, t, p) events is counted: windows containing fewer than `min_events = 500` events are discarded, while those meeting or exceeding this threshold proceed to voxelization as valid training samples. This filter is a structural necessity for SNN training stability:

- **Prevention of dead neurons:** SNNs are metabolic models: they require energy (spikes) to learn. Training on near-empty windows rewards the optimiser for suppressing all firing, driving synaptic weights so low that even high-density frames containing real vehicles fail to trigger a spike response Eshraghian et al. 2023.
- **Firing Rate Regularization (FRR) stability:** The FRR loss penalises deviation from a target firing rate of 20% Deng et al. 2023; Yan et al. 2022. Presenting near-empty inputs forces the regularisation term to dominate the total loss, as it attempts to generate spikes in the absence of any meaningful input signal.
- **Gradient quality:** A forward pass producing no spikes yields a zero surrogate gradient Emre O Neftci, Mostafa and Zenke 2019a. Training on empty windows is therefore mathematically degenerate since no weight update occurs, so High Performance Computing (HPC) cycles are consumed without any learning signal.
- **Background activity suppression:** Event cameras produce a baseline level of Background Activity (BA) noise from transistor mismatch and thermal effects

Gallego et al. 2022. A window containing less than 500 events is statistically likely to consist almost entirely of noise; the `min_events` threshold therefore functions as a signal intensity floor, directing the model’s attention exclusively towards windows in which a meaningful scene change has occurred.

1.3.2 Signal Conditioning

The first step of the `REPLICIA` pipeline is to apply a series of signal conditioning operations to the raw event data to improve the Signal-to-Noise Ratio (SNR), which yields several benefits:

- **Prevention of LIF neuron saturation:**

Without input conditioning, high-magnitude or noisy voxel values can drive every neuron in the initial stem layer to fire continuously and flood the network with uninformative spikes. By normalising the voxel tensor to the range $[0, 1]$, the input magnitude is constrained such that only neurons encoding genuine structural features, such as vehicle edges and motion boundaries, exceed their firing threshold, preserving the sparsity and selectivity of the deeper backbone stages Eshraghian et al. 2023.

- **Improved objectness discrimination:** Background artifacts inherent to High Definition (HD) event cameras, such as flickering illumination and sensor noise, generate erraneous event clusters that appear similar to small objects. By rejecting these low-activity events during the Background Activity Filtering (BAF) denoising step, the network is not required to expend representational capacity on learning to suppress noise, and can instead focus on learning the geometric structure of real vehicles Eshraghian et al. 2023; Roy, Jaiswal and Panda 2019a.

- **Reduced synaptic operations and neuromorphic energy efficiency:** Since SNNs consume energy only upon spike emission, every erraneous background event that is suppressed during preprocessing directly reduces the number of Synaptic Operations (SOPs) at inference time. This establishes a direct link between preprocessing quality and the power efficiency of the architecture when deployed on neuromorphic hardware Roy, Jaiswal and Panda 2019a.

The Signal Conditioning segment of the `REPLICIA` pipeline was lifted directly from the original ETrAM study, since it was designed specifically to suit the characteristics of the dataset Verma et al. 2024.

1.3.3 Spatiotemporal Voxelization

To make the asynchronous event data compatible with neural network layers, a Voxelization process is performed to transform the continuous stream into a structured tensor. This process consists of three core operations:

- **Temporal Binning:** The total duration of an event sample (e.g., a 100ms window) is divided into 10 discrete time bins ($T = 10$).
- **Polarity Splitting:** Events are separated into two distinct channels based on their polarity ($C = 2$), representing positive and negative brightness changes.
- **Event Accumulation (Histogramming):** Within each discrete temporal bin and polarity channel, the individual asynchronous events are aggregated at their respective spatial coordinates (x, y) . Specifically, the value of each “pixel” in the resulting spatial grid becomes the total count of events (of that specific polarity) that occurred at that exact location during that fraction of time. This process effectively converts the sparse, continuous point-cloud of events into a sequence of dense 2D frames—or event histograms—which standard convolutional and spiking layers can natively process.

This accumulation process preserves the temporal distribution of the events while generating a dense 4D spatiotemporal tensor of shape $[10, 2, 720, 1280]$ (representing Time $\times$ Channels $\times$ Height $\times$ Width).

1.3.4 Downsampling & Normalization

Following spatiotemporal voxelization, the voxelized data undergoes a dual-phase refinement consisting of spatial downsampling and amplitude normalization. This is the final polishing step of the pipeline, transforming the high-resolution event counts into a stabilized, low-memory format that the neural backbones can process efficiently.

Spatial Downsampling (16-fold Data Reduction)

The generated high-definition voxel grid (1280×720) is subjected to a 4×4 Max-Pooling operation, reducing the spatial resolution to 320×180 . This 16-fold reduction in data volume is critical for managing the high-dimensional state-space of the SNN backbone. Because SNNs must store neuron membrane potentials across time, an HD model would require massive amounts of GPU memory. By aggressively reducing

the spatial dimensions, we ensure the backbone scalability, making it computationally feasible to train the deep 4-block architecture over extended epochs.

Furthermore, utilizing Max-Pooling (as opposed to Average-Pooling) acts as a feature preserver: it retains the strongest event signal within each 4×4 area, helping the network maintain the crisp, high-contrast edges of the vehicles even at a lower resolution.

It should be noted that the downsampling factor $ds = 4$ was heavily influenced by the HPC cluster’s GPU memory constraints; without using $ds = 4$, the data would not fit into the GPU memory, or would require a batch size of 1, making training infeasible within the time constraints of this project.

Max-Normalization (Input Calibration)

Subsequently, max-normalization is performed to scale the raw event accumulation counts into a continuous $[0, 1]$ interval, by dividing every pixel in the temporal volume by the single highest event count found within that 10-bin voxel.

This normalization serves as a fundamental safeguard against neural saturation. LIF nodes have fixed firing thresholds (in this architecture, $V_{th} = 0.2$) - if a raw, unscaled event count of 50 were fed into the network, the first-layer neurons would fire instantaneously and repeatedly, effectively “screaming” and drowning out all other spatiotemporal information. By enforcing input calibration, we ensure that the input magnitudes are calibrated relative to the LIF thresholds, facilitating stable surrogate gradient propagation during the 150-epoch training cycle.

1.3.5 Multi-Object Label Alignment

The final stage of the pipeline implements a Multi-Object Label Alignment strategy, which is fundamental to the architecture’s multi-object detection capabilities. This stage transforms individual, asynchronous bounding box annotations into structured, scale-independent target tensors suitable for the network’s multi-box regression head.

Timestamp-Based Grouping (Unified Scene Representation)

Instead of creating isolated training samples for every individual vehicle, the pipeline parses the ground truth annotations and groups all vehicles that exist at the exact same microsecond (t). This process merges individual detections into a Unified Scene Representation, which is crucial because it eliminates “label noise”; presenting the

network with identical visual input but different single-car targets across multiple samples would force the model to average its predictions, resulting in blurry, low-confidence bounding boxes with poor IoU scores.

Fixed-Capacity Padding

Deep learning models inherently require consistent tensor dimensions for batched training. To accommodate scenes with varying numbers of vehicles, we enforce a Fixed-Capacity Interface of exactly 5 bounding boxes per frame. If a scene contains 2 vehicles, the remaining 3 slots are populated with “empty” zero-padded boxes. If a frame contains no vehicles, 5 empty boxes are provided. This ensures every target label is a consistent $[5, 5]$ tensor. The limit of 5 boxes was empirically selected based on an analysis of the ETrAM validation set, where the vast majority of frames contain 3 or fewer vehicles, making 5 a safe upper bound for this specific traffic monitoring environment.

Binary Masking

To enable the network to distinguish between genuine vehicle targets and zero-padded instances, the tensor utilizes a Binary Masking dimension. For every box, the target tensor stores five values: $[Center_X, Center_Y, Width, Height, Mask]$.

- For a real object, the mask value is set to 1.0.
- For a padded or empty box, the mask value is set to 0.0.

This mask acts as the objectness confidence target. It is utilized by the Multi-Box Loss function to selectively apply coordinate gradients only to valid targets, while simultaneously regularizing the model’s confidence head to output 0.0 for empty regions.

Normalization to Sensor Space

Finally, all spatial parameters (center coordinates, width, and height) are originally provided in raw pixel values (e.g., $x = 842$). These are normalized by dividing by the raw sensor width (1280) and height (720), scaling all coordinates into the continuous range $[0.0, 1.0]$. This provides a Scale-Independent Signal, ensuring the regression head can learn the underlying geometry robustly, regardless of the spatial downsampling factors applied earlier in the pipeline.

1.3.6 Chunking for High-Throughput I/O

To optimize throughput on the HPC cluster, the precomputed dataset was encapsulated into a High-Throughput Chunking architecture. Without this optimization, a 150-epoch training run on the high-definition event data would likely take days due to severe I/O bottlenecks.

The Chunking Mechanism

Rather than saving 10,000 individual files representing every single frame, the pipeline bundles the data into large, contiguous blocks, or “chunks” (e.g., 250 or 500 samples per file), where the chunk size is defined by the memory constraints of the environment used for training. Each chunk is saved as a single, compressed PyTorch (.pt) file containing two massive tensors: the spatiotemporal voxels and their corresponding aligned targets. To manage this structure, a global `manifest.json` file acts as a map, recording exactly which sample index resides in which chunk and specifying its byte-offset.

Mitigating HPC Bottlenecks

This chunking strategy was an absolute necessity for efficient execution on the cluster, addressing three primary hardware constraints:

- **Solving the “Small File Problem” (Metadata Latency):** HPC systems typically utilize a Network Attached Storage (NAS/NFS) architecture. Opening a file on a network drive incurs high *Metadata Latency*. If the GPU were forced to open 10,000 independent files per epoch, it would spend the vast majority of its time idling while waiting for the network to locate the next file. By encapsulating the data into chunks, the HPC only needs to open a fraction of the files (e.g., 40 files for 10,000 samples). This allows the system to perform massive *Sequential I/O* reads, which are orders of magnitude faster than random access reads on networked drives.
- **Bypassing CPU Precomputation Bottlenecks:** In deep learning pipelines, the CPU often acts as the data loader for the GPU. Preprocessing 1280×720 event data into voxel grids on-the-fly is highly computationally expensive.; by enforcing strict *Precomputation* and chunking everything prior to training, the training script can simply copy the structured tensors from disk directly into VRAM.

This ensures the CPU is no longer a bottleneck, maintaining *Maximum GPU Utilization* and enabling iteration speeds as low as 160ms/it.

- **Efficient Memory Caching (LRU):** The chunked architecture enabled the implementation of a custom `VoxelDiskDataset` featuring a Least Recently Used (LRU) memory cache. Because the data is bundled, the HPC can load entire blocks of 250 samples directly into its extensive RAM (e.g., 120GB). Consequently, during subsequent epochs, the model frequently bypasses disk I/O entirely, pulling chunks directly from RAM and making data loading nearly instantaneous.

By utilizing chunk-based sequential I/O and a memory-mapped caching layer, the *REPLICA* pipeline successfully bypassed the CPU-loading bottleneck, ensuring maximum GPU utilization and reducing the total 150-epoch training time from a projected several days to approximately six/ seven hours.

1.3.7 REPLICA Pipeline: Visualized

Figure 1.2 details the order of which the aforementioned preprocessing steps are applied to the raw event data.

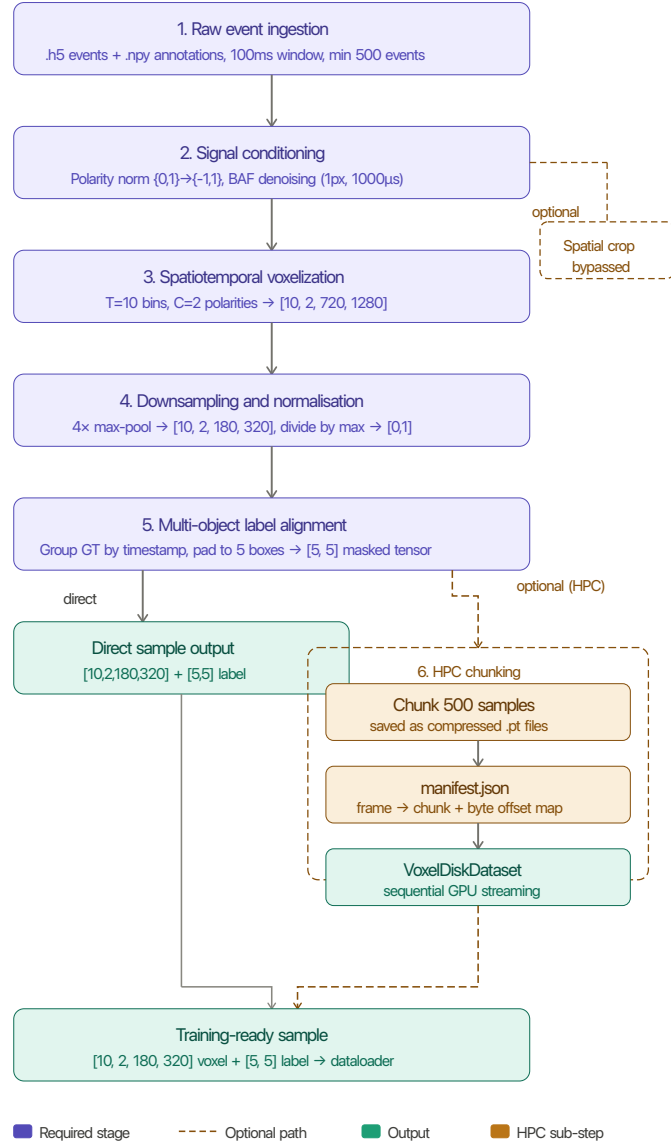


Figure 1.2: Overview of the spatiotemporal preprocessing pipeline applied to the ETrAM dataset. Raw events are voxelized into a 4D tensor, spatially downsampled, and aligned with multi-object ground truth labels to produce a final sample of shape [10, 2, 180, 320] with a corresponding label tensor of shape [5, 5].

1.4 Spiking Neural Network (SNN) Architecture

The core focus, effort and innovation of this project is the development and optimization of the `SNN_MAX_REPLICA_SE` architecture. This model is explicitly designed to process high-definition neuromorphic event data while adhering to the computational paradigms of SNNs. By prioritizing event-driven, binary communication, the architecture aims to drastically reduce both the dynamic power consumption and the computational complexity compared to traditional deep learning models, making it superior for power and time constrained environments.

1.4.1 Spatiotemporal Input & Multi-Step Execution

The raw ETraM event data is initially preprocessed to a spatial resolution of 640×360 with $T = 10$ discrete temporal bins. Consequently, the network ingests a massive, high-dimensional spatiotemporal tensor of shape $[B, 10, 2, 360, 640]$ (Batch, Time, Polarity, Height, Width).

Given the high spatial dimensionality and the iterative nature of spiking neurons, native Python loop execution over the time dimension proved computationally restrictive. To overcome this bottleneck, the entire architecture was refactored to utilize the SpikingJelly framework’s optimized Multi-Step (`step_mode='m'`) execution paradigm Fang, Chen et al. 2023; instead of iterating through the 10 time bins sequentially, the entire spatiotemporal tensor is passed directly into custom-compiled CUDA kernels. This low-level optimization enables high-speed, parallelized execution of the temporal dynamics, making the extensive benchmarking of this HD dataset feasible on local GPU hardware, and reducing overall training time from approximately 16 hours to just 7.

1.4.2 Neuron Dynamics: LIF

At the foundation of the architecture lies the LIF neuron model, which governs the temporal dynamics and binary spike generation of the network. Unlike traditional Artificial Neural Networks (ANNs) that utilize continuous activation functions such as ReLU, the LIF node maintains an internal state: the membrane potential $V[t]$, which updates iteratively across discrete time steps $t \in [1, T]$.

The Forward Pass (Spike Generation)

The dynamics of the LIF neuron at a given spatial position and time step t are governed by the integration of incoming pre-synaptic spatial features $X[t]$ and an inherent leak factor τ . The hidden membrane potential $H[t]$ prior to spiking is mathematically defined as:

$$H[t] = V[t-1] + \frac{1}{\tau} (X[t] - (V[t-1] - V_{reset})) \quad (1.1)$$

When $H[t]$ exceeds a predefined threshold V_{th} , the neuron emits a binary spike $S[t] \in \{0, 1\}$ to the subsequent layer, governed by the Heaviside step function $\Theta(x)$:

$$S[t] = \Theta(H[t] - V_{th}) \quad (1.2)$$

Following a spike ($S[t] = 1$), the neuron undergoes a hard reset, instantly dropping its potential $V[t]$ to V_{reset} . If no spike is emitted ($S[t] = 0$), the potential naturally decays, ensuring that subsequent layers only perform computations (Synaptic Operations) upon receiving active events Sengupta et al. 2019. This “activation sparsity” is the primary driver of the SNNs exceptional energy efficiency Roy, Jaiswal and Panda 2019b.

The Backward Pass (Surrogate Gradients)

Training deep SNNs via backpropagation presents a fundamental mathematical challenge, since the Heaviside step function governing spike generation (Equation 2) is non-differentiable. Its derivative is zero everywhere except at the threshold V_{th} , where it is infinite. Using conventional backpropagation here would immediately result in zeroed gradients, halting the learning process - this is a phenomenon known as the “dead neuron” problem.

To resolve this, the network employs a Surrogate Gradient during the backward pass Emre O. Neftci, Mostafa and Zenke 2019b. While the forward pass remains strictly binary, the backward pass approximates the step function using a smooth, differentiable curve. This specific architecture uses the ArcTangent (ATan) function:

$$\frac{\partial S}{\partial H} \approx \frac{\alpha}{2 \left(1 + \left(\frac{\pi}{2} \alpha (H - V_{th}) \right)^2 \right)} \quad (1.3)$$

Where α is a hyperparameter controlling the steepness of the surrogate curve. This mathematical substitution allows the network to learn complex spatiotemporal representations from scratch while maintaining a purely binary forward execution.

1.4.3 Backbone: SEW-ResNet & Attention

To extract high-level semantic features and manage the computational complexity of the high-definition input, the architecture utilizes a deep residual backbone divided into five sequential stages, as outlined in Table 1.1.

Stage	Name	Resolution	Channels	Primary Operation
1	Initial Stem	90×160	64	Stride-2 Conv + LIF (Downsamples HD input)
2	Backbone Alpha	22×40	256	Blocks 1 & 2 (SEWResBlocks with Stride-2)
3	Attention Bridge	22×40	256	Squeeze-and-Excitation (SE) Global Firing Recalibration
4	High-Res Backbone	22×40	512	Blocks 3 & 4 (SEWResBlocks with Stride-1)
5	Temporal Fusion	22×40	512	Summation of spikes across all $T=10$ timesteps

Table 1.1: Detailed breakdown of the five sequential stages in the `SNN_MAX_REPLICA_SE` backbone.

Spatial Downsampling

As illustrated, the architecture implements an aggressive $16\times$ total spatial downsampling strategy to compress the initial 640×360 input. This is achieved through strided convolutions applied within the Initial Stem and Backbone Alpha, progressively reducing the spatial dimensions to a highly compressed 22×40 grid while expanding the channel width up to 512.

Spike-Element-Wise (SEW) Residual Blocks

Building deep SNNs presents significant challenges due to the “vanishing spike” problem inherent in standard residual connections. In a standard Residual Neural Network (ResNet), the residual addition occurs *before* the activation function. However, if we follow this standard and add two binary spike trains, the result is a non-binary, continuous value (e.g., $1 + 1 = 2$), which destroys the energy-efficient, event-driven nature of the network Fang, Yu et al. 2022.

To overcome this, the `SNN_MAX_REPLICA_SE` architecture adopts the SEW residual paradigm Fang, Yu et al. 2022, instantiated as `SEWResBlock` units throughout the backbone. Unlike standard Spiking ResNets, SEW ResBlocks apply the element-wise residual operation *after* the spiking activation function, guaranteeing that the output

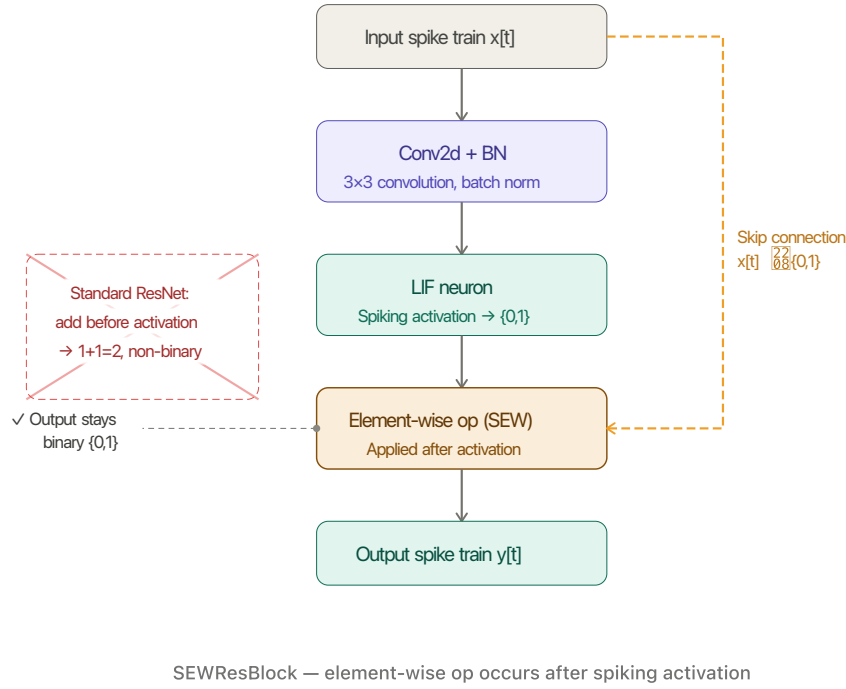


Figure 1.3: The SEW residual block (SEWResBlock) as proposed by Fang, Yu et al. 2022.

remains strictly binary. Fang et al. Fang, Yu et al. 2022 prove that this reordering is both necessary and sufficient to achieve identity mapping in a purely spike-based network, resolving the degradation problem without sacrificing neuromorphic compatibility. In this architecture, this property is exploited to stack 4 SEWResBlocks across the backbone stages (Stages 2 and 4 in 1.1), enabling representational depth that would otherwise cause spike degradation under a standard residual formulation.

Structural Attention: SE-Block

In deep learning, a bottleneck refers to a point in the neural network where the information flow is forced through its most compressed and abstract representation. In this SNN_MAX_REPLICA_SE architecture, the bottleneck exists at the transition point between stages 2 and 3 of the backbone, and represents two things:

1. **A Spatial Constraint (Resolution Limit):** At this stage, the original 180×320 spatiotemporal input has been downsampled by a factor of 8x via the strided convolutions in the stem and early residual block, resulting in a highly compressed $22 \times$

40 feature map. This spatial bottleneck is critical because each remaining "pixel" must now represent a large 8×8 receptive field of the original HD sensor data, forcing the network to discard fine-grained geometric redundancy while retaining the core structural boundaries of multiple detected objects.

2. **A Semantic Shift (High-Level Abstraction):** This juncture marks the definitive pivot where the network transitions from extracting low-level geometric primitives, such as spike-edge orientation and motion direction, to synthesizing complex semantic concepts. Without an explicit selection mechanism, the network at this depth is susceptible to noise and artifacts from the HD event stream, which can blur the distinction between legitimate targets and background interference.

To both mitigate and solve these bottleneck constraints, a Squeeze-and-Excitation (SE) mechanism is introduced in Stage 3 of the backbone, sitting directly at the transition point Hu et al. 2019. The SE block operates by performing a global average pooling of the 256 feature channels (the "Squeeze" phase), followed by two fully connected layers that calculate a weighting factor for each channel (the "Excitation" phase). This effectively mitigates the spatial constraint by allowing the network to incorporate global context into its local representations, compensating for the loss of resolution.

When applied in the context of an SNN, this acts as a dynamic firing modulator: the backbone's activations are re-calibrated by looking at the global context of the entire frame, magnifying the most informative channels whilst suppressing those containing artifacts. By applying this recalibration, the SE block ensures that only the most relevant "Objectness" (the probability that a specific region of an image contains a generic object of interest rather than background noise) features are passed into the final high-capacity blocks, effectively mitigating the semantic ambiguity before the final bounding box regression.

1.4.4 Firing Rate Regularization (FRR)

A common failure mode in deep SNN training is network silencing (zero spikes emitted) or over-saturation (firing continuously at every timestep), both of which halt gradient flow and destroy energy efficiency. To constrain the network, a custom Firing Rate Regularization (FRR) Metabolic Penalty was integrated into the training pipeline.

The FRR applies a squared error loss to the network's average firing rate:

$$L_{frr} = \lambda \cdot (Rate_{avg} - Target)^2 \quad (1.4)$$

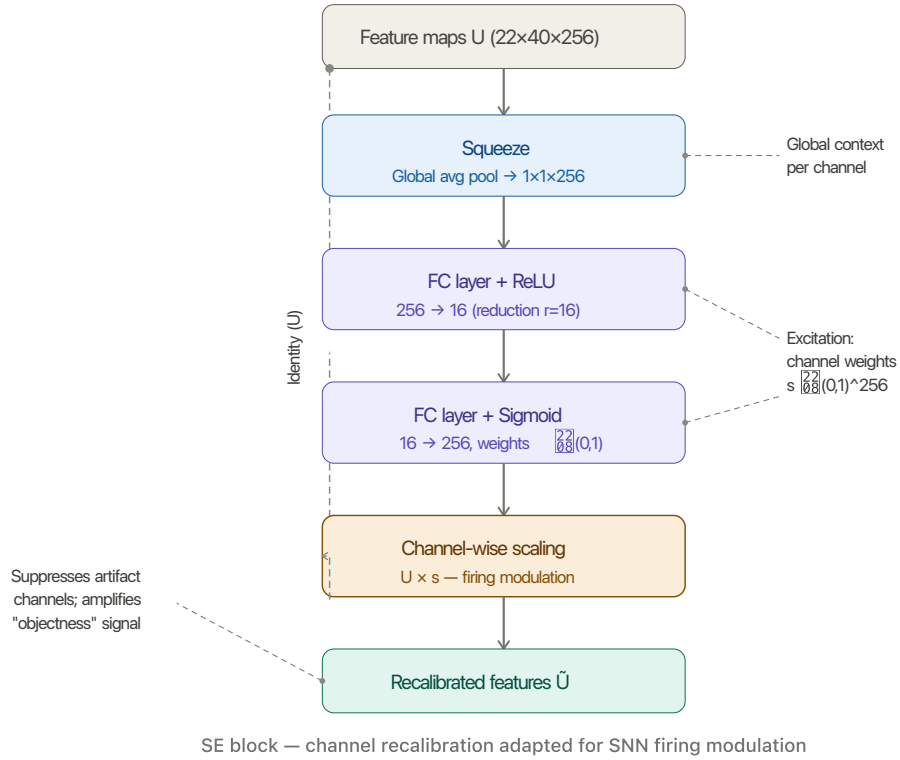


Figure 1.4: The Squeeze-and-Excitation (SE) block Hu et al. 2019 as applied in Stage 3 of the SNN_MAX_REPLICA_SE backbone ($22 \times 40 \times 256$).

This biologically inspired metabolic constraint Roy, Jaiswal and Panda 2019a forces the 11.7M parameter backbone to maintain a selective but informative firing rate (e.g., $Target = 20\%$) across its hidden layers. Penalising derivation from a target firing rate is an established technique for preserving activation sparsity in directly trained SNNs Deng et al. 2023; Yan et al. 2022. Adding this penalty to the model’s loss ensures high information flow while strictly preserving the activation sparsity required for efficient neuromorphic deployment.

1.4.5 Multi-Box Regression Head

Following the final backbone stage, the abstract spatiotemporal feature maps are fed into a unified detection head. To process the spatial information effectively and mitigate spatial ambiguities observed during the iterative design process (detailed in Section 1.6), the head utilizes an 8×8 Adaptive Average Pooling mechanism.

This pooling operation aggregates the features into 64 distinct spatial receptive

blocks. The pooled features are then mapped via final convolutional and linear layers to an output tensor of shape $[B, 5, 5]$, where B represents the batch size. The final two dimensions encode the top 5 predicted bounding boxes globally per image, with each box defined by a feature tuple $\mathcal{F} = \{x, y, w, h, c\}$. Here, (x, y, w, h) represent the normalized spatial coordinates and dimensions of the bounding box, and c denotes the object confidence logit. This formulation allows the network to predict multiple vehicles simultaneously, which are subsequently evaluated using a vectorized Multi-Box Loss function.

1.5 Convolutional Neural Network (CNN) Baseline

The Convolutional Neural Network (CNN) architecture `CNN_MAX_REPLICA_SE` was engineered as the primary baseline to evaluate the SNN against. It is a 11.73M parameter CNN specifically designed to mirror the SNN architecture, satisfying the ‘‘Architectural Parity’’ functional requirement specified in 1.1.1. This ensures that the only variable being tested in this comparative study is the neural activation mechanism (event-driven binary spikes versus continuous-valued floats), making the comparison immune to common criticisms regarding unmatched or unoptimized baselines.

1.5.1 Structural Configuration & Layer-Wise Mapping

The CNN mirrors the SNN’s configuration strictly 1:1, replacing every temporal and spiking component with a standard continuous equivalent. Table 1.2 outlines this direct mapping.

1.5.2 Input Strategy: Channel-Wise Temporal Integration

Unlike the SNN, which processes event data sequentially over discrete time steps T , the CNN baseline utilizes a flattened temporal input strategy. Instead of processing 10 timesteps sequentially, it concatenates the $T = 10$ bins (each containing 2 polarity channels) into a single input tensor of 20 channels, changing the input shape from $[B, 10, 2, H, W]$ to $[B, 20, H, W]$.

This approach allows the CNN to correlate events across the entire 100ms window in a single spatial pass. This channel-wise stacking of event bins is a standard and effective methodology for feeding asynchronous event-camera data into traditional, non-spiking CNN architectures Gehrig et al. 2019.

SNN Component	CNN Equivalent	Technical Purpose
Spatiotemporal Voxel $[B, 10, 2, H, W]$	Flattened Channel Input $[B, 20, H, W]$	Translates 10 time bins into a format the respective network can ingest.
LIF Neuron (Leaky Integrate-and-Fire)	ReLU Activation	Provides the nonlinearity; SNN uses discrete pulses, CNN uses continuous values.
Binary Spikes $\{0, 1\}$	Continuous Floats $[0, \infty)$	Represents the information “currency” being passed between layers.
Surrogate Gradient	Exact Derivative (Standard)	Enables backpropagation through the non-differentiable spiking threshold.
SEWResBlock (Spiking Shortcut)	Standard Residual Block	Implements identity mapping to prevent vanishing gradients during training.
Multi-Step Mode (<code>step_mode= 'm'</code>)	Single spatial pass	Processes the temporal window efficiently on the GPU.
Squeeze-and-Excitation (Spike-based)	Squeeze-and-Excitation (Float-based)	Recalibrates channel importance based on global frame context.
Temporal Summation ($\sum_{t=1}^T S_t$)	N/A (Direct Feature Pass)	Aggregates 100ms of evidence into a single map for the regression head.
Multi-Box Head (5 Boxes)	Multi-Box Head (5 Boxes)	Predicts coordinates and objectness scores for 5 targets simultaneously.
Synaptic Operations (SOPs)	Floating Point Ops (FLOPs)	The metric used to quantify and compare hardware energy efficiency.

Table 1.2: 1:1 Architectural Mapping (SNN vs. CNN)

1.5.3 Backbone: Identity Mapping & Residual Connections

The CNN backbone employs deep residual learning, utilizing four Residual Blocks that expand to 512 channels in the final stage. To satisfy the functional requirement of Architectural Parity, the spiking SEWResBlocks from the SNN were replaced with Standard Residual Blocks. The standard structure within these blocks is $\text{Conv} \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{Conv} \rightarrow \text{BatchNorm} + \text{Shortcut} \rightarrow \text{ReLU}$.

The inclusion of these skip connections (shortcuts) is vital for enabling the CNN network to avoid the same vanishing gradient problem seen in its SNN counterpart, and successfully learn complex vehicle geometries across the 320×180 field of view He et al. 2015.

1.5.4 SEWResBlock vs Standard ResBlock

The core mathematical difference between the two architectures lies in their summation logic and the resulting information flow across the shortcut connection:

- **Standard ResBlock (CNN):** Performs an accumulative floating-point addition ($y = f(x) + x$). The continuous intensity of the residual path is directly added to the identity path before the final ReLU activation. Here, the shortcut carries the *raw input features* forward. This mechanism allows the model to learn the “difference” (the residual) needed to improve the representation, focusing on information preservation He et al. 2015.

- **SEWResBlock (SNN):** Performs a relational element-wise spiking operation *after* the activation function. Because adding two binary spikes (e.g., $1 + 1 = 2$) produces a non-binary value, the SNN uses a spike-merging function (such as ADD or AND) to preserve the discrete $\{0, 1\}$ state. Crucially, the SNN shortcut carries the *membrane history* forward, prioritizing temporal integrity.

Table 1.3 summarizes these foundational differences in identity mapping.

Feature	Standard ResBlock (CNN)	SEWResBlock (SNN)
Summation	Accumulative ($y = f(x) + x$)	Relational (Spiking Element-Wise)
Output Type	Continuous Float $[0, \infty)$	Binary Spikes $\{0, 1\}$
Primary Benefit	Prevents vanishing gradients	Prevents spike vanishing
Mathematical Goal	Information preservation	Temporal integrity

Table 1.3: Comparison of identity mapping between Standard ResBlocks and SEWRes-Blocks.

This distinction is particularly relevant for high-speed neuromorphic data: while the Standard ResBlock accumulates continuous values that effectively blur fast movements into a single averaged float over the 100ms window, the SNN equivalent preserves exact spike timing, maintaining the temporal precision required for the regression head.

1.5.5 Attention Mechanism: SE-Block

The CNN network encounters the same informational bottlenecks as its SNN counterpart, detailed in Section 1.4.3. Consequently, it implements the same architectural solution via an SE Block Hu et al. 2019. However, the `CNN_MAX_REPLICA_SE` implements this mechanism in the continuous domain: as opposed to utilizing spike rates, the CNN employs a float-based approach.

During the ‘Squeeze’ segment, the CNN utilizes global average pooling to calculate the mean spatial intensity of features per channel, rather than calculating a temporal firing rate. Subsequently, during the ‘Excitation’ phase, the block outputs a learned coefficient that scales the input activations. This effectively recalibrates the channel-wise feature responses, as opposed to modulating discrete membrane potentials.

While the integration of an SE Block in a standard CNN improves feature representation, it does not confer the same critical hardware-level energy sparsity benefits as it does within an SNN. The primary advantage of the SE mechanism in this specific context is the enforcement of structural sparsity in the spiking domain. Nevertheless, to

strictly adhere to the Architectural Parity requirement established in Section 1.1.1, the continuous equivalent of this mechanism was preserved in the baseline.

1.5.6 The “Gold Standard” Upper Bound

By constructing this custom CNN mirror, the baseline serves as the upper theoretical bound of this specific architecture’s capability on the ETraM dataset. For example, if the CNN baseline achieves a Mean IoU of 48%, this represents the maximum performance extractable by this parameter configuration. If the corresponding SNN achieves 43% IoU, the 5% gap explicitly represents the “quantization cost” of switching from 32-bit continuous floats to 1-bit binary spikes Cordone, Miramond and Thierion 2022. Adhering to this methodology ensures that the evaluation of the neuromorphic paradigm is both fair and scientifically sound.

1.6 Architectural Evolution

The final **ULTRA** revision of the `MAX_REPLICA_SE` architecture was not the result of a single design phase, but rather the culmination of an iterative engineering process across four distinct generations. According to the project’s success criteria, the model required both high temporal precision and robust spatial localization. This section documents the empirical obstacles encountered during development, and the actionable insights that led to the final optimized configuration.

1.6.1 Phase 0: The Spatial Cropping Pitfall (Early Prototyping)

Prior to the implementation of downsampling in the `REPLICA` pipeline, early prototyping explored the use of aggressive spatial cropping to bypass the computational costs of high-definition event processing.

Architectural Intent & Technical Logic

The primary objective of this phase was to establish a high-resolution localized baseline by cropping the raw 1280×720 ETraM sensor data into a 224×224 patch, as illustrated in Figure 1.2; these patches were deterministically anchored to the centroid of the ground-truth vehicle annotation.

The technical logic behind this approach was rooted in improving the SNR at the network’s input. By utilizing *localized foveation*, we hypothesized that the initial stem layers would receive a significantly higher frequency of spikes per pixel for the target object. This increased spike density was intended to facilitate the extraction of fine-grained geometric features which are often attenuated or lost during global spatial downsampling, whilst also making it easier for the network to learn the features of vehicles by enlarging and zooming in on them. Additionally, since the localized patches occupied a smaller memory footprint, they could be processed without the need for aggressive pooling, and preserve the original sensor resolution.

Empirical Obstacle: Constant Target Bias

During initial training cycles, the model achieved near-instantaneous convergence and exceptional accuracy metrics. However, a critical failure in the methodology was identified during validation: the model had failed to learn any meaningful feature representation.

Because the pipeline utilized deterministic centroid-anchored cropping, the target vehicle occupied the geometric center of every training sample. Consequently, the SNN succumbed to *Constant Target Bias*—a severe form of spatial overfitting where the regression head ignores the visual event stream and instead learns to output a constant coordinate vector of approximately $[0.5, 0.5, w, h]$. This eliminated the localization signal entirely; the model stopped “searching” for the vehicle and simply memorized the center coordinates. Such a model is fundamentally useless for real-world traffic monitoring where vehicle positions are stochastically distributed across the full field of view, and are not pre-known to use spatial cropping before inference..

Actionable Insight

This phase provided a critical methodological lesson: for a spiking architecture to learn robust spatial localization and objectness, it must be exposed to targets moving dynamically relative to the sensor boundaries within a full-frame (or semi-full-frame) field of view. This insight led to the total abandonment of centered cropping in favor of the global $4\times$ spatial downsampling strategy utilized in the final *REPLICA* pipeline, ensuring that the model learned to correlate visual spike patterns with absolute spatial coordinates.

1.6.2 Generation 1: The Multi-Box Paradigm Shift

Following the resolution of the spatial cropping bias, the architecture progressed to full-frame processing. However, this transition immediately exposed fundamental limitations in the network’s predictive capabilities regarding multi-object scenes.

Architectural Intent & Technical Logic

The primary objective of this phase was to develop a *scene-aware detector* capable of identifying multiple vehicles simultaneously within the ETraM traffic environment. To achieve this, the network’s output topology underwent a paradigm shift, transitioning from a simplistic single-box regression head (which output a single $[x, y, w, h]$ vector) to a Multi-Box Regression Head with a fixed-capacity output of $[B, 5, 5]$, where there are 5 features $(x, y, w, h, mask)$ and 5 bounding boxes predicted per inference. The technical logic dictated that the network required a mechanism to independently evaluate distinct spatial regions. To facilitate this, each of the five predicted boxes was augmented with a fifth dimension: an “Objectness” or Confidence Score. This structural addition allowed the model to not only regress bounding box coordinates but also to gate its own predictions, learning to suppress background artifacts with low confidence while maintaining high confidence for legitimate structural targets.

Empirical Obstacle: Single-Target Ambiguity & Label Noise

Initial prototypes utilizing the standard single-box regression head proved fundamentally incompatible with real-world traffic scenes. When multiple vehicles occupied the field of view simultaneously, the model faced a logical contradiction.

For instance, if a scene contained both Car A and Car B, the SNN backbone would naturally extract features for both. However, if the training label only provided the coordinates for Car A, the loss function would heavily penalize the network’s correct detection of Car B as an error, introducing severe label noise. Over thousands of iterations, the model attempted to minimize this ambiguous loss by mathematically averaging the positions of all visible vehicles. This catastrophic failure mode resulted in a predicted bounding box that hovered uselessly in the empty space between vehicles, or a singular “super-box” attempting to encompass the entire roadway, leading to a collapse in IoU performance.

Actionable Insight: The Vectorized Matching Loss

This phase highlighted that resolving spatial ambiguity required a sophisticated mechanism to handle the *Assignment Problem*—the mathematical challenge of determining which of the 5 predicted boxes should be compared against which ground truth vehicle.

The actionable insight was the development of a Vectorized Multi-Box Loss. This computational innovation calculates a 3D matrix of all possible IoU overlaps between the 5 predictions and the M ground truth objects in every batch. Utilizing greedy matching, the loss function identifies the best match for each real vehicle in the scene. Only this matched predicted box receives a *coordinate gradient*, signaling it to move closer to the target.

Crucially, all unmatched predicted boxes are assigned a target confidence of 0.0, acting as a metabolic penalty for false positives. This forces the model to learn that predicting a vehicle where none exists carries a heavy mathematical penalty, effectively cleaning up the SNN’s vision and reducing the Ghost Detections prevalent in early iterations.

By transitioning to a multi-box architecture, we moved the SNN from a simplistic regression model to a sophisticated scene-aware detector, and introduced the Multi Box Label Alignment step shown in figure 1.2 into the REPLICA pre-processing pipeline. This shift allowed the network to resolve the spatial ambiguity of multi-vehicle traffic, transforming the contradictory Label Noise of previous versions into a structured multi-target learning signal. This evolution was essential for achieving stable convergence on the high-definition, high-density event streams of the ETrAM dataset.

1.6.3 Generation 2: The Expanded Spatial Head (7.00M Parameter MAX Generation)

While Generation 1 successfully established the multi-box logic, empirical validation revealed a persistent failure mode characterized by ghost Detections and overlapping predictions.

Architectural Intent & Empirical Obstacle

The primary objective of this phase was to resolve *Information Congestion* within the regression head. Generation 1 architecture utilized an Adaptive Average Pooling layer of 4×4 , compressing the entire field of view into just 16 spatial blocks before entering the final linear layers.

This severe spatial bottleneck meant that when vehicles were in close proximity, or situated near clusters of sensor artifacts, their respective spiking features were averaged into an identical 4×4 block. The regression head lacked the spatial memory required to distinguish valid target spikes from background noise. Consequently, the model frequently outputted overlapping bounding boxes for a single vehicle or failed to suppress high-confidence noise, resulting in an inflated average of 2.50 detections per frame - significantly higher than the ground truth average of 2.0.

Actionable Insight: Quadrupling Spatial Memory

To resolve this congestion, the Adaptive Pooling resolution was expanded from 4×4 to an 8×8 grid, quadrupling the available spatial memory to 64 distinct blocks, providing the linear regression head with a substantially finer structural map of the backbone's activations.

This architectural refinement enabled localized Object-Noise Discrimination: with 64 distinct blocks, the model could successfully isolate the structured spike trains of a vehicle from the unstructured noise of adjacent sensor artifacts. This resulted in dramatic Confidence Polarization: max confidence scores for real objects surged to 0.99, while background noise was successfully pushed toward 0.0.

Consequently, the average detections per frame dropped to 2.06, nearly perfectly matching the ground truth. However, while the model's selectivity improved drastically, the Mean IoU plateaued at 41%. This confirmed that while the regression head now possessed sufficient capacity, the backbone's heavily downsampled resolution remained the fundamental bottleneck preventing higher geometric precision.

1.6.4 Generation 3: The High-Resolution Backbone (11.7M Parameter ULTRA Generation)

Following the expansion of the spatial head in Generation 2, the model demonstrated excellent selectivity but remained computationally constrained by its geometric precision.

Architectural Intent & Empirical Obstacle

The primary objective of this final phase was to break the Precision Ceiling established in Generation 2. Despite the regression head's capacity to isolate targets, the Mean IoU remained stagnant at 41%.

The fundamental obstacle was the severe spatial compression occurring within the backbone. By the final layer, the feature map had been downsampled to an 11×20 resolution. At this scale, a single pixel in the final feature map represented a massive 16×16 area of the original HD sensor. Consequently, if a vehicle's bounding box edge shifted by several pixels in the real world, the network was entirely blind to the change; the MAX generation simply lacked the geometric detail required to draw tight, pixel-perfect bounding boxes.

Furthermore, an analysis of the network's information flow revealed that the Generation 2 backbone was suffering from neural silencing, exhibiting an average firing rate of $< 1.5\%$. This extreme sparsity starved the regression head of the necessary temporal evidence required for high-confidence predictions.

Actionable Insight: Semantic Expansion & Metabolic Awakening

To overcome this precision ceiling, a multi-faceted architectural overhaul was implemented to create the final ULTRA configuration:

- **Resolution Boost & Semantic Depth:** To provide the necessary geometric detail, the convolution stride in Block 3 was reduced to $s = 1$. This effectively halted the downsampling earlier in the network, doubling the final feature map resolution to 22×40 (quadrupling the available spatial cells to 880). To handle the increased complexity and high-frequency sensor artifacts exposed by this higher resolution, the backbone's semantic was expanded via the addition of a 4th SEWResBlock (512 channels). This allowed the network to think deeper about the structural composition of the vehicle.
- **Metabolic Regularization:** To wake up the silent backbone and maximize information flow, an aggressive metabolic strategy was deployed. The LIF firing threshold was lowered drastically ($V_{th} = 0.5 \rightarrow 0.2$), making it significantly easier for neurons to activate. Concurrently, the FRR penalty multiplier (λ) was quintupled ($1.0 \rightarrow 5.0$) with a target rate of 20%. This forced the network to generate a dense, high-frequency stream of geometric data, providing the regression head with the robust temporal evidence needed to sense the exact boundaries of the vehicles.
- **Hardware Optimization:** Expanding the parameter count to 11.7M and quadrupling the spatial resolution exponentially increased the computational load. To

prevent catastrophic latency penalties, the architecture was refactored to utilize SpikingJelly’s `step_mode='m'` (*Fused-Kernel Optimization*) Fang, Chen et al. 2023, which bypassed the Python interpreter to execute the temporal dimension ($T = 10$) directly on the GPU. This was coupled with Automatic Mixed Precision (AMP) utilizing 16-bit tensor cores, yielding a $\sim 1.8\times$ training speedup and making the ULTRA configuration computationally viable.

Generation 3 successfully orchestrated the transition from basic structural detection to high-fidelity geometric precision. By halting the spatial degradation and enforcing strict metabolic regularization, the ULTRA model demonstrated that SNNs can achieve precise localization on HD event data without the severe latency penalties typically associated with deep spiking architectures.

1.6.5 Generations: Summarized

To summarize, Table 1.4 shows the evolution of the `DetectorNX` architecture across the three generations.

Generation	Primary Archi- tectural Shift	Params	Main Strength	Final Limitation
Gen 1: Base	Multi-Box Head [$B, 5, 5$]	5.43M	Handled multi-vehicle scenes simultaneously	High false positives in low-density regions (“Ghosting”)
Gen 2: Expanded	8×8 Spatial Pooling Head	7.00M	Solved object-noise ambiguity through improved spatial aggregation	41% IoU ceiling due to insufficient spatial resolution
Gen 3: Ultra	22×40 Resolution with additional SEWResBlock (Stage 4)	11.72M	High-fidelity edge precision through full-resolution backbone processing	Higher computational metabolic cost in SOPs

Table 1.4: Generational evolution of the `DetectorNX` architecture.

1.7 Experimental Setup & Benchmarking

Having established the theoretical design and optimization of the `DetectorNX` architecture, both the SNN and its CNN counterpart, the final step is to formulate a rigorous evaluation strategy to empirically validate its performance against the continuous CNN baseline. In accordance with the project’s primary objectives, the evaluation cannot be limited to a singular metric of bounding box accuracy; rather, it must holistically assess the neuromorphic paradigm. To achieve this, a comprehensive benchmarking suite was designed (as mentioned in Figure 1.1) comprising five distinct experimental protocols:

- Dynamic Power Estimation
- Quantization Error Analysis (Accuracy vs. Object Size)
- Confidence Calibration
- Temporal Evidence Accumulation
- Real-time Inference latency
- Theoretical Energy Consumption
- Global Evaluator: Mean/Median IoU, Recall, and Average Detections Per Frame

This multi-faceted methodology ensures that the comparative findings are statistically robust and firmly grounded in the established literature surrounding event-based vision.

1.7.1 Dynamic Power Estimation

A fundamental objective of this research is to quantify the energy efficiency gains achieved by transitioning from a continuous CNN to a bio-plausible SNN. Traditional CNN architectures are characterized by static power consumption; they perform a dense, fixed number of Multiply-Accumulate (MAC) operations for every forward pass, regardless of the information density within the input scene. In contrast, SNNs leverage event-driven dynamics, where metabolic energy is consumed only when a neuron reaches its firing threshold and generates a discrete spike.

The Dynamic Power experiment investigates the reactive scalability of the architecture by mapping instantaneous energy consumption against the temporal density of the

input stream. This analysis serves to validate the event-driven nature of the SNN, demonstrating that its metabolic cost is a dynamic function of activation sparsity—mimicking biological neural efficiency by consuming energy only in response to informative stimuli.

CNN Power Estimation

In a standard CNN, the total energy per frame (E_{CNN}) is calculated by multiplying the total number of Floating-Point Operations (FLOPs) by the cost of a 32-bit MAC (E_{MAC}):

$$E_{CNN} = \sum_{l=1}^L (FLOPs_l \times E_{MAC}) \quad (1.5)$$

Where $E_{MAC} = 4.6$ pJ per 32-bit floating point operation, as referenced in Table 1.5 Horowitz 2014.

The number of FLOPs is the total number of MACs operations required to produce the output feature map; in a CNN, this number is static, regardless of what the image contains. The FLOPs of a CNN can be calculated as follows:

$$FLOPs_l = (H_{out} \times W_{out}) \times (K \times K) \times C_{in} \times C_{out} \quad (1.6)$$

Where $(H_{out} \times W_{out})$ is the number of pixels in the output frame, $(K \times K)$ is the size of the convolutional kernel, C_{in} is the number of input channels, and C_{out} is the number of output channels.

SNN Power Estimation

For the spiking architecture, the calculation is scaled by the activation sparsity (ζ) of each layer. The metabolic cost of the spiking backbone can be quantified via the SOP metric; unlike the static complexity of the CNN baseline, the SOP count for a given layer is a dynamic function, which can be calculated as follows:

$$SOPs_l = FLOPs_l \times \zeta \times T \quad (1.7)$$

where ζ_l represents the average firing rate of layer l , T is the number of temporal simulation steps ($T = 10$).

In order to obtain ζ_l , a specialized `SpikeRateMonitor` tool was developed. The tool attaches a hook to every LIF inside the SNN backbone, which listens to the layer during

the forward pass and intercepts the binary spike tensor before it moves to the next layer. For every layer l , the SNN produces a spike tensor S_l with the shape $[T, B, C, H, W]$, where $T = 10$ is the number of timesteps, B is the batch size, and C, H, W are the spatial dimensions. Since a spike is binary (1 for fire, 0 for rest), ζ can be calculated by taking the global mean of the entire tensor:

$$\zeta_l = \frac{1}{N} \sum_{i=1}^N S_{l,i} \quad (1.8)$$

To calculate Dynamic Power, ζ was calculated by running inference over all validation samples and taking the mean firing rate per layer across all samples in the dataset.

Because SNNs utilize binary activations, the expensive multiplication part of the MAC operation is omitted, leaving only a Conditional Addition (AC) of the weight to the membrane potential. The total energy per frame (E_{SNN}) is defined as:

$$E_{SNN} = \sum_{l=1}^L (SOP_{S_l} \times E_{AC}) \quad (1.9)$$

where $E_{AC} = 0.9$ pJ is the energy cost of a single floating-point addition, as referenced in Table 1.5 Horowitz 2014.

1.7.2 Quantization Error Analysis (Accuracy vs. Object Size)

The Quantization Error Analysis experiment investigates the discretization threshold of the SNN architecture and quantifies the extent to which the 1-bit binary nature of spiking activations limits the model's geometric acuity. Unlike the continuous CNN baseline, which utilizes 32-bit floating-point activations to represent spatial coordinates, the SNN is constrained by the finite temporal resolution of its simulation window ($T = 10$).

Core Metrics and Scale Partitioning

The primary performance metric for this experiment is the Mean Intersection Over Union (mIoU), which quantifies the overlap between the predicted bounding box (P) and the ground truth (GT). To analyze the impact of scale on regression precision, the validation set is partitioned into three distinct categories based on the absolute pixel area (A_{px}) of the ground truth objects:

- **Small:** Objects in the bottom 33rd percentile of area (typically distant vehicles or small artifacts).
- **Medium:** Objects in the middle 33rd percentile.
- **Large:** Objects in the top 33rd percentile (vehicles in close proximity to the sensor).

Mathematical Framework

The categorization and evaluation are governed by two fundamental geometric formulas. First, the pixel area is calculated by denormalizing the ground truth dimensions relative to the raw sensor resolution:

$$A_{px} = (W_{norm} \cdot W_{sensor}) \times (H_{norm} \cdot H_{sensor}) \quad (1.10)$$

where $W_{sensor} = 1280$ and $H_{sensor} = 720$. Subsequently, the precision of the model’s spatial regression is quantified via the IoU formula:

$$IoU = \frac{\text{Area}(P \cap GT)}{\text{Area}(P \cup GT)} \quad (1.11)$$

Hypothesis: The Scale-Dependent Accuracy Gap

We hypothesize a significant scale-dependent accuracy gap between the two architectures. For large-scale objects, the SNN is expected to achieve parity with the CNN baseline: because the coordinate values are high, the minor fluctuations inherent in binary spike-trains represent a negligible percentage of the total object area, allowing for stable regression.

Conversely, for small-scale objects, a substantial performance divergence is anticipated. In an SNN with $T = 10$, a neuron can only represent $T + 1$ discrete states ($0/10, 1/10, \dots, 10/10$). As documented by Cordone, Miramond and Thierion 2022, this rounding error, or quantization penalty, is massive relative to the dimensions of small vehicles. If a target requires a precise coordinate gradient (e.g., 0.154) for accurate localization, the SNN is physically forced to discretize this value, establishing a fundamental “resolution floor” that limits mIoU on small features. While the use of SEWResBlock architectures improves signal flow Fang, Yu et al. 2022, it does not inherently resolve this quantization of the final regression output, which is further exacerbated by the low event counts typical of small objects in event-based vision Gallego

et al. 2022.

1.7.3 Confidence Calibration

The Confidence Calibration evaluates predictive reliability by investigating the calibration gap between spiking and continuous architectures. Modern CNNs are prone to systematic overconfidence, frequently assigning high objectness scores to inaccurate detections Chuan Guo et al. 2017. Conversely, the temporal dynamics of SNNs may offer superior uncertainty quantification, providing a more robust reliability profile for safety-critical automotive tasks Skatchkovsky, Jang and Simeone 2022.

Methodology: Reliability Diagrams

To quantify calibration, we generate a Reliability Diagram mapping predicted confidence (C) against actual regression accuracy (IoU) for every validation detection. In a perfectly calibrated system, these values align along the identity line ($C = IoU$). Clumping in the high-confidence, low-accuracy quadrant indicates an overconfidence bias, where the model fails to signal its own predictive errors.

Hypothesis: Spikes as Uncertainty Sensors

The CNN baseline is hypothesized to exhibit significant miscalibration due to its continuous activation functions and exact gradient descent, which tend to overfit the confidence head. The SNN is expected to demonstrate a more linear, calibrated trend. In the spiking domain, low-quality features generate fewer spikes, naturally leading to lower membrane potentials in the detection head. This mechanism allows the SNN to function as an implicit uncertainty sensor, where discrete spike rates effectively gate the output confidence logit relative to the strength of the temporal evidence.

1.7.4 Temporal Evidence Accumulation

The Temporal Evidence Accumulation experiment investigates the sequential decision-making dynamics inherent to the SNN architecture. Unlike the CNN baseline, which performs a single spatial pass over the fully integrated 100ms voxel, the SNN processes the event stream iteratively across $T = 10$ timesteps. This property makes SNNs intrinsically suited to low-latency spatiotemporal integration — the capacity to produce meaningful predictions before the full observation window has elapsed Pfeiffer and

Pfeil 2018. The experiment therefore examines whether this theoretical advantage manifests empirically: specifically, whether the model can produce a valid bounding box regression before the 100ms window is complete.

In the context of vehicle object detection, this is a highly significant experiment — for example, if the SNN can accurately show a bounding box at $t = 5$ (50ms) instead of the full $t = 10$ (100ms) as the CNN would, this would lead to a 50% reduction in detection time. In a time-sensitive domain such as vehicle and pedestrian detection, such advantages could be the difference between a crash and a safe stop.

Methodology: Temporal Snapshots

To evaluate the evolution of predictions, we extract intermediate outputs at three specific checkpoints: $t = 2$ (20ms), $t = 5$ (50ms), and $t = 10$ (100ms). For each snapshot, the candidate bounding box coordinates and instantaneous confidence scores are recorded to observe the *hunting behavior* of the regression head: the observable process where the networks’ predicted bounding box moves, vibrates or searches for the object’s true edges before finally settling into a stable, high-precision detection. This process is driven by the discrete LIF dynamics:

$$v_i[t] = v_i[t - 1] + \frac{1}{\tau} \left(\sum_j w_{ij} s_j[t] - (v_i[t - 1] - V_{reset}) \right) \quad (1.12)$$

where $v_i[t]$ represents the accumulated evidence (membrane potential) and $\sum w_{ij} s_j[t]$ represents the incoming observations (spikes). This recursive filtering allows the network to update its belief about vehicle localization with every incoming pulse of data.

Hypothesis: Bounding Box Sharpening

We hypothesize a phenomenon of *Bounding Box Sharpening* over time. At $t = 20$ ms, we anticipate high coordinate variance, as the majority of LIF neurons will not have accumulated sufficient membrane potential to cross the firing threshold ($V_{th} = 0.2$) within the first two of ten timesteps. As t progresses toward 100ms, continued spike train integration allows stochastic membrane noise to average out, consolidating responses to the vehicle’s geometric edges into a stable coordinate estimate.

This progressive refinement is analogous in spirit to the “rapid categorisation” phenomenon documented in the human visual system, wherein early feedforward spikes

drive coarse semantic decisions that sharpen as further neural evidence accumulates Thorpe, Fize and Marlot 1996. While Thorpe et al. address categorical recognition rather than coordinate regression, the underlying principle — that sparse early spike activity produces meaningful but imprecise representations which sharpen over time — is consistent with the anticipated behaviour of the SNN backbone.

1.7.5 Real-Time Latency Benchmark

This experiment quantifies the forward-pass inference speed of both architectures on conventional Graphics Processing Unit (GPU) hardware at batch size $B = 1$, simulating a live single-stream event camera deployment. Latency (ms/f) and throughput (FPS = 1000/ms/f) are measured using `torch.cuda.Event` timing to capture true GPU execution time, bypassing Central Processing Unit (CPU)-to-GPU communication overhead.

Hypothesis

The CNN baseline is expected to exhibit lower latency, as GPUs are optimised for the dense matrix multiplications that underpin its single flattened spatial pass. The SNN must instead simulate LIF membrane dynamics iteratively:

$$v[t] = v[t - 1] + x[t] - \beta \cdot v[t - 1] \quad (1.13)$$

This constitutes a hardware-mismatch penalty: the latency advantage of SNNs only fully materialises on asynchronous neuromorphic hardware, not on synchronous GPU accelerators Davies et al. 2018; Blouw et al. 2019. The experiment aims to quantify this penalty and assess whether `step_mode='m'` parallelisation Fang, Chen et al. 2023 brings the SNN within a $2\times$ – $3\times$ latency margin of the CNN.

Limitations

The measured GPU latency figures cannot be extrapolated to neuromorphic silicon. The A100’s 312 TFLOPS (FP16) characterises dense matrix throughput, which cannot equally be judged against asynchronous spike-based execution. While Loihi benchmarks report up to $10\times$ lower latency over embedded GPUs for batch-size-1 inference Davies et al. 2018, these figures are task-specific; direct deployment on neuromorphic hardware is identified as future work.

1.7.6 Theoretical Energy Consumption

The Theoretical Energy Consumption is very similar in nature to the Dynamic Power experiment discussed in Section 1.7.1. It uses the same principles and formulae for SOPs and FLOPs, but differs in the final calculation: as supposed to calculating the dynamic power consumption per frame, this experiment aims to find the average power consumption per frame across the entire dataset, and extrapolates for multiple hardware platforms:

$$\bar{E}_{SNN} = \sum_{l=1}^L (SOPs_l \times E_{chip}) \quad (1.14)$$

$$\bar{E}_{CNN} = \sum_{l=1}^L (FLOPs_l \times E_{MAC}) \quad (1.15)$$

The calculation mirrors Equation 1.9, however $\bar{E}_{SNN}$ replaces E_{AC} for E_{chip} , which is the power consumed per SOP for different chipsets as depicted in Table 1.5.

Subsequently, the total percentage energy saving is formulated using the following equation:

$$\text{Energy Savings} = \left(1 - \frac{\bar{E}_{SNN}}{\bar{E}_{CNN}}\right) \times 100\% \quad (1.16)$$

Energy Constants Per Operation (E_{chip}) Across Multiple Platforms

Architecture / Platform	Operation Type	Energy (E_{chip})	Primary Citation
CNN Baseline	MAC	4.6 pJ	Horowitz 2014
Theoretical Silicon Limit	AC	0.9 pJ	Horowitz 2014
Intel Loihi 1	SOP	23.6 pJ	Davies et al. 2018
IBM TrueNorth	SOP	26.0 pJ	Merolla et al. 2014

Table 1.5: Energy constants per operation (E_{chip}) for each architecture and platform.

Hypothesis

We hypothesize that the SNN will demonstrate a metabolic advantage of at least one order of magnitude ($> 90\%$) over the CNN baseline, predicated on two complementary properties of the $\bar{E}_{SNN}$ formulation:

1. **Activation sparsity:** By exploiting the sparsity of the ETraM event stream, the backbone is expected to maintain $\zeta < 5\%$, bypassing the majority of computations

that the CNN performs densely regardless of scene activity.

2. **Mathematical simplification:** Replacing MAC operations with binary AC operations yields a $5\times$ reduction in per-operation energy cost (0.9 pJ vs 4.6 pJ), which is expected to provide a substantial efficiency dividend even when accounting for the $T = 10$ temporal overhead Horowitz 2014.

When extrapolated to physical neuromorphic platforms, the SNN’s event-driven resting state is expected to preserve a power advantage over synchronous GPU accelerators despite the higher per-SOP costs of Intel Loihi (23.6 pJ) and IBM TrueNorth (26.0 pJ) documented in Table 1.5 Davies et al. 2018; Merolla et al. 2014.

1.7.7 Global Performance Evaluator

The Global Performance Evaluator serves as the primary comparative evaluation between the SNN and CNN architectures, aggregating performance across the full 200-sample validation set. Unlike the preceding targeted experiments, this benchmark establishes the *Performance-Efficiency Frontier*: quantifying the accuracy penalty incurred by transitioning to discrete spiking dynamics, whilst validating whether the multi-box refinements and others in the ULTRA revision of `DetectorNX` mentioned in Section 1.6 are sufficient to resolve scene-level detection ambiguity.

Metrics

In this experiment, four complementary metrics are evaluated:

- **mIoU:** measures geometric precision between predicted and ground truth bounding boxes.
- **Recall:** measures the proportion of ground truth objects successfully localised at confidence threshold.

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